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GR-CLASS: Comprehensive Technical & Functional SRS

1. System Blueprint

Core Tech Stack

- **Backend:** Node.js (v20+), Express.js (v5)
- **Database:** MySQL 8.0 (Sequelize ORM)
- **Caching:** Redis
- **Cloud:** AWS (S3 for documents, SES for Email)
- **Mobile Support:** Optimized API for offline-first surveyor apps.

2. User Roles & Permission Matrix

The system uses **Role-Based Access Control (RBAC)**. Each role has a specific "Data Scope".

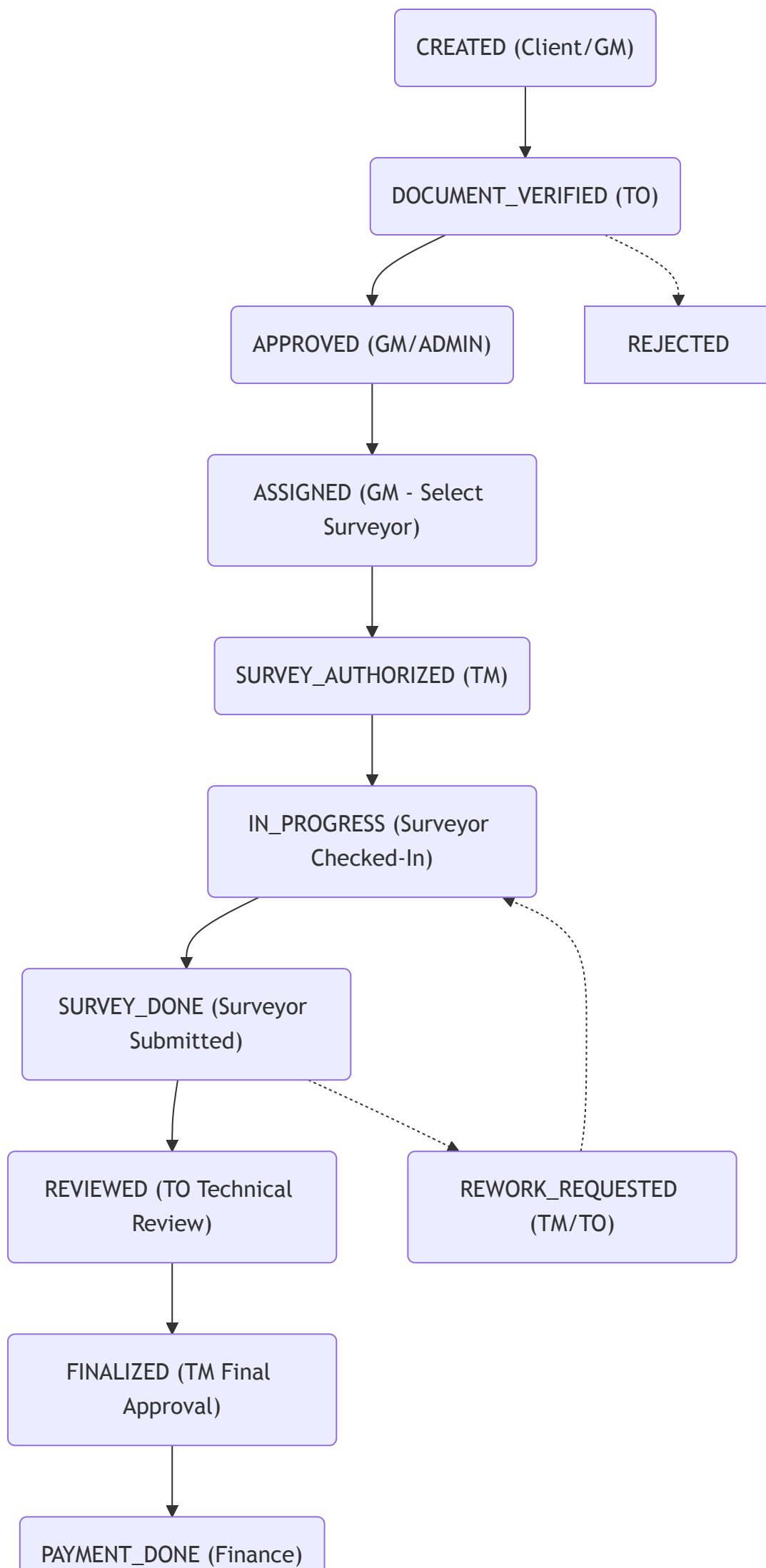
Role	Meaning	Responsibilities	Data Scope
ADMIN	Administrator	Total system control, user management, system config.	Global
GM	Gen. Manager	Business flows, Client/Surveyor onboarding, Job Approval.	Global
TM	Tech. Manager	Technical oversight, Final Survey Approval, Cert Issuance.	Global
TO	Tech. Officer	Document Verification , Technical review of checklists.	Global
SURVEYOR	Inspector	Performing vessel surveys, uploading reports.	Assigned Jobs Only
CLIENT	Ship Owner	Vessel management, requesting surveys, cert downloads.	Own Organization Only

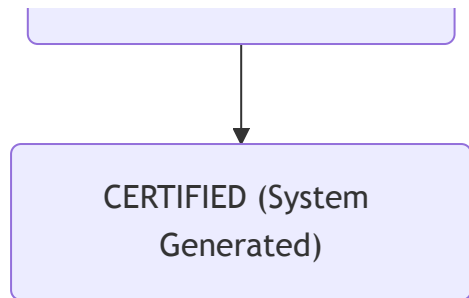
[!NOTE] **FLAG_ADMIN** is a reserved scope for external regulatory oversight and is currently managed via specialized Flag State modules, but it is not a direct system login role in the current DB schema.

3. The "Engine": Workflow Cycles

3.1 Job Request Lifecycle

This is the primary state machine of the application.





3.2 Activity & Change Request Funnels

Before a Job is created, or when data needs updating, these auxiliary workflows apply:

Activity Request (Sales/Commercial): DRAFT → PENDING → APPROVED → CONVERTED_TO_JOB (or REJECTED)

Change Request (Audit Trail): PENDING (User Request) → APPROVED (Admin/GM sets new value) or REJECTED

3.2 Surveyor Field Workflow (Hybrid Digital-Physical)

To satisfy maritime legal requirements for wet signatures on complex layouts, the system uses a hybrid approach:

1. **Check-in:** Surveyor hits **POST /surveys/start**. Start GPS and timestamp recorded.
2. **Digital Summary:** Surveyor marks high-level sections (e.g., Hull, Engine) as Pass/Fail in the app for instant reporting.
3. **Audit Flow:** Surveyor downloads master blank templates (**template_files** - JSON array), prints/signs them, and uploads scanned copies (**signed_checklist_files** - JSON array) per job.
4. **Signature:** Surveyor and Vessel Master sign and stamp the physical papers.
5. **Digitization:** Surveyor uploads the scanned/photographed **Signed Checklists** to the job (**PUT /surveys/jobs/:id/signed-checklist**).
6. **Check-out:** Surveyor submits the final survey report.

4. Module API Reference

4.1 Authentication (/auth)

- **POST /auth/login**: Identity verification.
- **POST /auth/refresh-token**: Renew short-lived access tokens.
- **POST /auth/logout**: Revoke tokens and clear cookies.
- **POST /auth/forgot-password**: Email-based reset.

4.2 Job Management (/jobs)

Method	Endpoint	Description	Roles
GET	/	List jobs with status/client filters.	All (Scoped)
POST	/	Request a new survey/cert.	CLIENT, GM, ADMIN
PUT	/:id/verify-documents	Mark docs as checked.	TO
PUT	/:id/approve-request	Approve for surveyor assignment.	GM
PUT	/:id/assign	Link a Surveyor to the job.	GM
PUT	/:id/authorize-survey	Unlocks survey flow for surveyor.	TM
PUT	/:id/review	Mark technical review complete.	TO
PUT	/:id/finalize	Final signature before billing.	TM, GM
PUT	/:id/priority	Update job urgency (URGENT/NORMAL).	GM, TM

4.3 Survey Operations (/surveys)

- **POST /surveys/start**: Record check-in coordinates.
- **POST /surveys/jobs/:id/proof**: Upload evidence photo (S3).
- **POST /surveys/jobs/:id/location**: Stream GPS pings during inspection.

- **POST** `/surveys/jobs/:id/sync`: Bulk upload offline-cached data.
- **PUT** `/surveys/jobs/:id/finalize`: TM command to lock the report.

4.4 Certification Engine (`/certificates`)

- **GET** `/certificates/types`: List available certs (e.g., Annual, Interim).
- **POST** `/certificates/`: Generate a draft from job data.
- **POST** `/certificates/:id/issue`: Converts draft to official Signed PDF.
- **GET** `/certificates/verify/:number`: Public endpoint for QR verification.
- **PUT** `/certificates/:id/renew`: Start renewal workflow (extends expiry).

4.5 Client & Vessel Management (`/vessels`, `/clients`)

- **GET** `/vessels`: List vessels (Filter by Flag, Age, Client).
- **POST** `/vessels`: Register new vessel info + IMO number.
- **GET** `/clients/:id`: View company details & primary contacts.
- **PUT** `/clients/:id/status`: Suspend/Activate organization.

4.6 Supporting Modules

- **Non-Conformities (NC)**: **POST** `/nc` (Create deficiency), **PUT** `/nc/:id/close` (Verify rectification).
- **Checklists**: **GET** `/checklist-templates` (Static definitions) vs **PUT** `/checklists/jobs/:id` (Surveyor input).
- **Dashboard**: **GET** `/dashboard` (Role-specific counters for Pending Tasks, Expiring Certs, Revenue).
- **Support**: **POST** `/support/tickets` (Issue reporting for clients/surveyors).
- **Notifications**: **GET** `/notifications` (In-app alerts for job status changes).

4.7 Commercial & Audit Trail Modules

- **Activity Requests**: **POST** `/activity-requests` (Client inquiry for non-standard services).

- **Change Requests:** `POST /change-requests` (Formal vessel/cert data correction with admin approval).
- **Approvals:** `POST /approvals` (Generalized multi-step approval engine for complex entities).

4.8 Technical Oversight (TOCA)

- **Transfer of Class (TOCA):** `POST /toca` (Manage vessel records when gaining or losing class).
- **Incidents:** `POST /incidents` (Report vessel casualties or detentions).

4.9 Collaboration & Communication

- **Job Messaging:** `POST /jobs/:id/messages` (Contextual chat between Client/Surveyor/Admin).
- **Internal Notes:** `POST /jobs/:id/notes` (Private management comments visible to TM/GM).
- **Documents Module:** `GET /documents/:entityType/:id` (Universal file access across the system).

4.10 Website & CMS

- **Static Content:** `GET /site-static/:slug` (Manage Terms of Service, FAQ, Policy text).
- **Video CMS:** `POST /website/videos` (Manage portfolio/training videos).
- **Newsletter:** `POST /website/newsletter/subscribe` (Marketing funnel).

5. Technical Constraints for Frontend Implementation

5.1 Success/Error Handling

The backend always responds with a standardized wrapper:

- **Status 200/201:** Success. `json.success = true`.
- **Status 400:** Validation Error (Check `json.message` for Joi details).
- **Status 401/403:** Auth/RBAC issues.
- **Status 404:** Item not found.

5.2 Document Uploads

The backend uses **Pre-signed URLs** for maximum performance.

1. Frontend requests `GET /api/v1/certificates/upload-url`.
2. Backend returns an S3 `uploadUrl` and a `fileKey`.
3. Frontend performs `PUT` to AWS S3 directly.
4. Frontend notifies Backend with `fileKey` in the final `POST`.

5.3 Offline Synchronization

Surveyors often work in areas with 0 connectivity. The Frontend should:

- Cache Checklist data locally (IndexedDB/Redux Persist).
 - Store GPS points with timestamps.
 - Use `POST /surveys/jobs/:id/sync` to replay the entire session once online.
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6. Role-Based Navigation Mapping (Checklist)

Client Panel

- ☐ View Dashboard (My Vessels, Active Jobs).
- ☐ Request New Job.
- ☐ Upload Vessel Supporting Docs.
- ☐ Download Issued Certificates.

Admin/Manager Panel (GM/TM)

- ☐ List all Client Requests (New).
- ☐ Assign Surveyors (Map integration recommended).
- ☐ Review Surveyor Submissions (Photo gallery view).
- ☐ Audit Logs & Report Builder.

Surveyor Panel (The "Field Unit")

- ☐ View Assigned Jobs Calendar.
- ☐ Interactive Checklists (Toggle switch for Pass/Fail).
- ☐ Camera Integration (Evidence).

7. Complete Route Dictionary (Exhaustive List)

Module	Method	Path	Roles
Auth	POST	/auth/login	Public
	POST	/auth/refresh-token	Public
	POST	/auth/logout	All
	POST	/auth/forgot-password	Public
	POST	/auth/reset-password	Public
	POST	/auth/change-password	All
Users	GET	/users/me	All
	GET	/users	ADMIN
	POST	/users	ADMIN
	PUT	/users/:id	ADMIN
	PUT	/users/:id/status	ADMIN
	PUT	/users/profile-pic	All
	PUT	/users/fcm-token	All
Vessels	GET	/vessels	All (Scoped)

Module	Method	Path	Roles
	POST	/vessels	ADMIN, GM, TM
	GET	/vessels/:id	All (Scoped)
	PUT	/vessels/:id	ADMIN, GM, TM
	GET	/vessels/client/:clientId	ADMIN, GM, TM
Jobs	GET	/jobs	All (Scoped)
	POST	/jobs	CLIENT, ADMIN, GM
	GET	/jobs/:id	All (Scoped)
	PUT	/jobs/:id/verify-documents	TO
	PUT	/jobs/:id/approve-request	GM
	PUT	/jobs/:id/assign	GM
	PUT	/jobs/:id/authorize-survey	TM, ADMIN
	PUT	/jobs/:id/review	TO
	PUT	/jobs/:id/finalize	TM, GM
	PUT	/jobs/:id/reschedule	GM
	PUT	/jobs/:id/priority	GM, TM
	GET	/jobs/:id/history	ADMIN, GM, TM, TO
	POST	/jobs/:id/notes	ADMIN, GM, TM, TO
	POST	/jobs/:id/messages	All (Scoped)
	GET	/jobs/:id/eligible-surveyors	ADMIN, GM, TM
Surveys	POST	/surveys/start	SURVEYOR
	POST	/surveys/jobs/:id/proof	SURVEYOR

Module	Method	Path	Roles
	POST	/surveys/jobs/:id/location	SURVEYOR
	POST	/surveys/jobs/:id/sync	SURVEYOR
	PUT	/surveys/jobs/:id/signed-checklist	SURVEYOR
	POST	/surveys/	SURVEYOR (Submit)
	PUT	/surveys/jobs/:id/finalize	TM
	PUT	/surveys/jobs/:id/rework	GM, TM
	POST	/surveys/jobs/:id/statement/issue	TM
Checklists	GET	/checklist-templates	ADMIN, GM, TM
	POST	/checklist-templates	ADMIN
	PUT	/checklist-templates/:id	ADMIN
	GET	/checklists/jobs/:id	All (Scoped)
	PUT	/checklists/jobs/:id	SURVEYOR
Certificates	GET	/certificates/types	All (Scoped)
	POST	/certificates	TM, GM
	POST	/certificates/:id/issue	GM
	GET	/certificates/:id/download	All (Scoped)
	PUT	/certificates/:id/suspend	TM
	PUT	/certificates/:id/revoke	TM
	PUT	/certificates/:id/renew	TM
	POST	/certificates/:id/reissue	TM
	GET	/certificates/verify/:num	Public
Activity	POST	/activity-requests	CLIENT, ADMIN, GM, TM
	GET	/activity-requests	All (Scoped)

Module	Method	Path	Roles
Change Req	PUT	/activity-requests/:id/status	ADMIN, GM, TM
	POST	/change-requests	All (Scoped)
	PUT	/change-requests/:id/approve	ADMIN, GM
NC	POST	/nc	SURVEYOR, TO
	PUT	/nc/:id/close	TO, TM
	GET	/nc/job/:jobId	All (Scoped)
Incidents	POST	/incidents	CLIENT, ADMIN, GM, TM
	GET	/incidents	All (Scoped)
TOCA	POST	/toca	TM
	GET	/toca	ADMIN, GM, TM
Clients	GET	/clients	ADMIN, GM, TM, TO
	POST	/clients	ADMIN, GM, TM
	PUT	/clients/:id/status	ADMIN, GM
Documents	GET	/documents/:type/:id	All (Scoped)
	POST	/documents/register	All
	GET	/documents/get-upload-url	All
Payments	GET	/payments	ADMIN, GM, TM
	POST	/payments	ADMIN, CLIENT
System	GET	/system/health	All

Module	Method	Path	Roles
	GET	/system/audit-logs	ADMIN
	GET	/system/metrics	ADMIN
Website	POST	/website/newsletter/subscribe	Public
	POST	/website/videos	ADMIN
	GET	/site-static/:slug	Public
Feedback	POST	/feedback	CLIENT
	GET	/feedback/public	Public
	PATCH	/feedback/:id/visibility	ADMIN
Search	GET	/search	All
Dashboard	GET	/dashboard	All